

Aim:-

To design and set up the active first order filters using op amp.

Apparatus Required :-

Op-amp, resistors, capacitors, breadboard, CRO, function generator and power supplies.

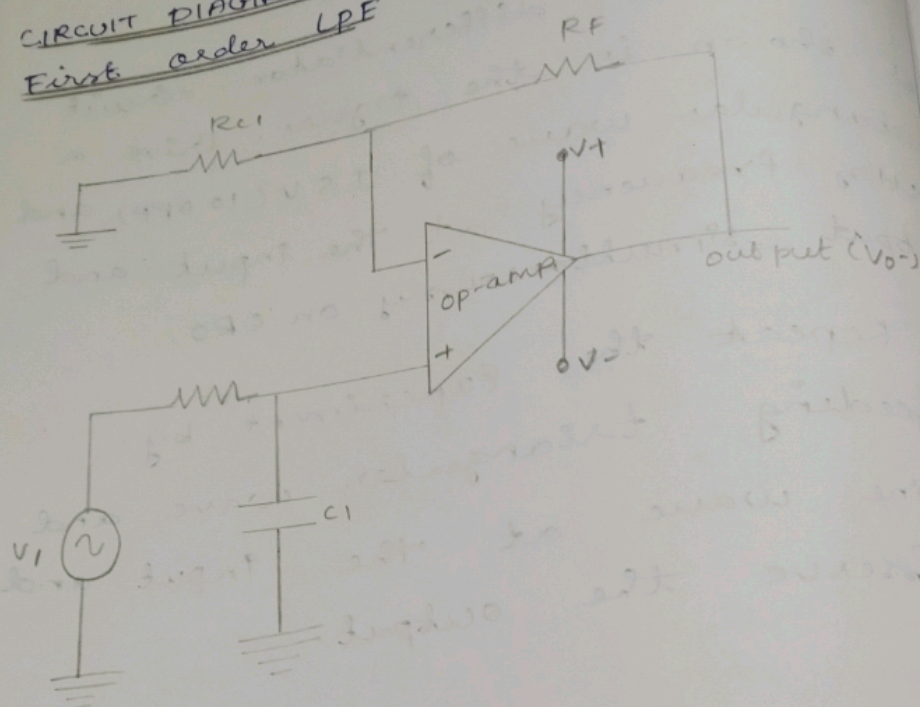
Design:-

$$F_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 2.2 \times 10^3 \times 0.0001 \times 10^{-6}}$$
$$= 720 \text{ Hz}$$

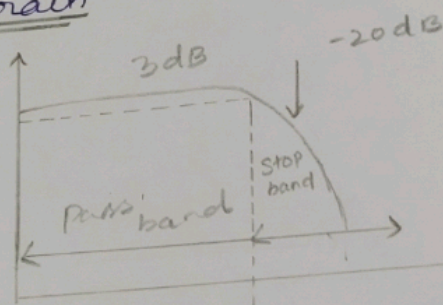
Procedure:-

1. Assemble the first-order active filter circuit on the breadboard.
2. connect IC 741 to a $\pm 15 \text{VDC}$ supply.
3. Connect the function generator to the input of the filter.
4. Apply a sinusoidal input, for eg 1V_{rms}
5. Initially set the input frequency well below the calculated cut-off frequency.
6. Observe V_{in} and V_{out} on the CRO/DSO.

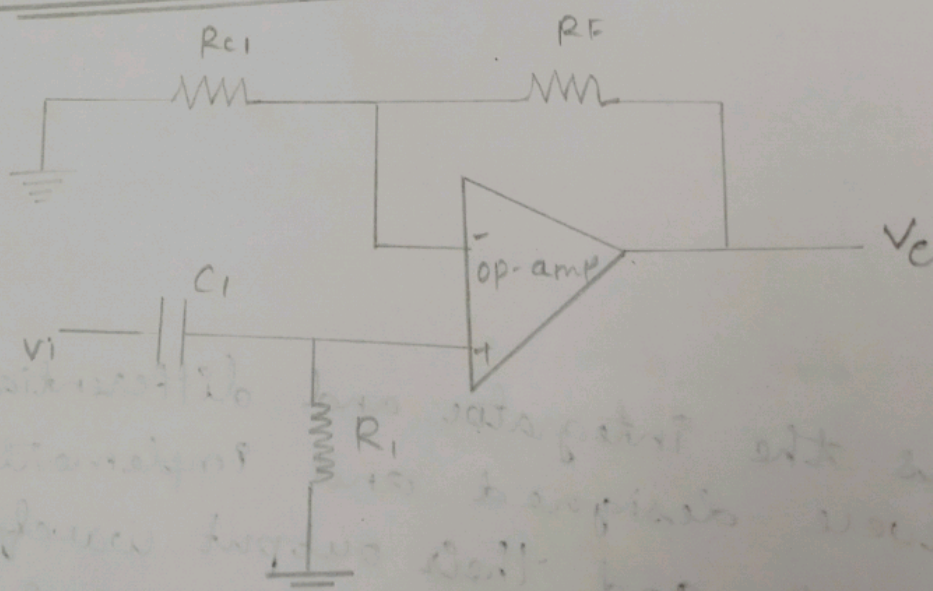
CIRCUIT DIAGRAM First order LPE



Voltage Gain



First order HPF



EXP No: 6
DATE: 20/08/2026

Aim:-

To design
First order

Apparatus R

op-amp
breadboard
and power

Design:-

$F_c =$

Procedure

1. Ass
for

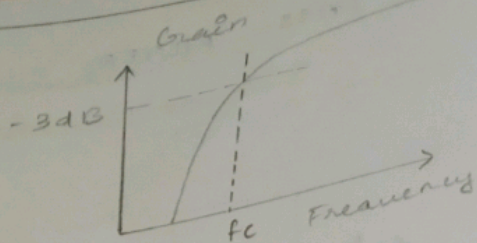
2.

3.

4.

5.

6.



OBSERVATION TABLE:-

S.NO	Frequency (Hz)	Vin (V)	Vout (V)	Gain A	Gain dB $20 \log(A)$
1.	30 Hz	1V	1V	1	0
2.	50 Hz	1V	1V	1	0
3.	80 Hz	1V	1V	1	0
4.	100 Hz	1V	1V	1	0
5.	150 Hz	1V	1V	1	0
6.	500 Hz	1V	500 mV	500	-6.02
7.	1K Hz	1V	400 mV	400	-7.9
8.	2K Hz	1V	200 mV	200	-13.9
9.	3K Hz	1V	100 mV	100	-20

7. Measure different
8. Calculate decimal
9. Increase through cut-off
10. Record
11. Identify
12. Plot
- 13.

7. Measure the output voltage at different input frequencies.
8. Calculate the voltage given in decimal.
9. Increase the frequency gradually through and beyond the expected cut-off frequency.
10. Record the output voltage at each frequency.
11. Identify the frequency-response curve of gain (dB) versus frequency.
12. Plot on the frequency-response curve of gain (dB) versus frequency.
13. Compare the experimentally obtained cut-off frequency with theoretical value.

Result:-

Thus designed the first order filters using op-amp and obtained the results as cut-off frequency of low pass filter

